

1. Local Projections: Step-by-Step Implementation (Jordà, 2005)

This section describes a transparent, step-by-step procedure for estimating regime-conditional impulse responses using local projections following Jordà (2005, <https://www.aeaweb.org/articles?id=10.1257>). The approach is especially convenient in our setting because it (i) accommodates regime dependence via threshold indicators and (ii) allows us to include a rich set of controls without estimating a full VAR.

1.0.1. Step 1: Define the shock

Let ΔD_t denote the weekly change in the broad U.S. dollar index. We interpret $\Delta D_t > 0$ as a dollar appreciation shock.

1.0.2. Step 2: Choose horizons

Fix a maximum horizon H (in weeks). We estimate responses for horizons

$$h = 0, 1, 2, \dots, H.$$

In the baseline results, we set $H = 8$, corresponding to an eight-week response window.

1.0.3. Step 3: Define the outcome variable

Let $x_{n,t}$ denote the JPY–USD cross-currency basis at maturity n . For each horizon h , we use the cumulative change in the basis from $t - 1$ to $t + h$ as the dependent variable:

$$y_{n,t}^{(h)} \equiv x_{n,t+h} - x_{n,t-1}. \quad (1)$$

This construction yields an impulse response interpretation for the coefficients estimated at each horizon h .

1.0.4. Step 4: Define dollar regimes (threshold indicator)

Let D_t denote the level of the broad dollar index and let $\hat{D}^\tau$ be the estimated threshold. Define the regime indicators

$$\mathbf{1}_t^L \equiv \mathbf{1}(D_t < \hat{D}^\tau), \quad \mathbf{1}_t^H \equiv \mathbf{1}(D_t \geq \hat{D}^\tau), \quad (2)$$

so that $\mathbf{1}_t^L + \mathbf{1}_t^H = 1$ for all t .

1.0.5. Step 5: Specify the local projection regression (one regression per horizon)

For each maturity n and horizon h , we estimate the following OLS regression:

$$x_{n,t+h} - x_{n,t-1} = \beta_n^L(h) \Delta D_t \mathbf{1}_t^L + \beta_n^H(h) \Delta D_t \mathbf{1}_t^H + \Gamma_n(h)' Z_t + \alpha_n(h) + \varepsilon_{n,t+h}, \quad (3)$$

where Z_t is a vector of controls (e.g., ΔVIX_t , JPY/USD implied volatility, risk reversals, and U.S.–Japan yield and term spread differentials), $\alpha_n(h)$ is a horizon-specific intercept, and $\varepsilon_{n,t+h}$ is the regression residual.

1.0.6. Step 6: Interpret coefficients as impulse responses

The sequences $\{\beta_n^L(h)\}_{h=0}^H$ and $\{\beta_n^H(h)\}_{h=0}^H$ are the regime-conditional impulse responses of the basis to a one-unit dollar appreciation shock in the low- and high-dollar regimes, respectively. We summarize the state dependence by comparing these responses across horizons:

$$\beta_n^H(h) - \beta_n^L(h), \quad h = 0, 1, \dots, H.$$

1.0.7. Step 7: Use appropriate standard errors (overlapping horizons)

Because the dependent variable in (3) uses overlapping observations for $h \geq 1$, residuals are serially correlated. We therefore compute heteroskedasticity-and-autocorrelation-robust (HAC) standard errors using Newey–West adjustments. A common choice is to set the HAC lag length proportional to the horizon, e.g.,

$$\ell_h = h + 1, \tag{4}$$

though results are typically robust to reasonable alternative choices.

1.0.8. Step 8: Plot impulse response functions

For each maturity n , we plot $\beta_n^L(h)$ and $\beta_n^H(h)$ against h and report 95% confidence bands constructed. These figures visualize (i) the persistence of basis responses following a dollar shock and (ii) the degree of regime asymmetry.

1.0.9. Step 9: Optional scaling to one standard deviation shocks

For presentation, it is often useful to scale the responses to a one-standard-deviation dollar shock:

$$\tilde{\beta}_n^\theta(h) \equiv \beta_n^\theta(h) \cdot \sigma_{\Delta D}, \quad \theta \in \{L, H\},$$

where $\sigma_{\Delta D}$ is the sample standard deviation of ΔD_t .

Local projections estimate impulse responses by running separate regressions for each horizon h . In our regime-dependent specification, the dollar shock ΔD_t is interacted with threshold indicators to obtain distinct impulse responses in low- and high-dollar states and plot the estimated response paths across h .